Homework 5

Biostatistics 735
Due: November 17, 2026

1. **Clustering [25 points]**

   The `USArrests` data set is already in R. It contains, for each of the 50 US states in 1973, the number of arrests per 100,000 residents for murder, assault, and rape, and the percent of the population living in urban areas.

   (a) (3 pts) Standardize the four variables with `scale()`. Explain why this matters for clustering here. (Hint: look at the variances of the four variables before scaling.)

   (b) (6 pts) Apply **K-means** clustering to the standardized data. Choose the number of clusters using `fviz_nbclust()` from the `factoextra` package, with both the elbow (`method = "wss"`) and silhouette (`method = "silhouette"`) methods. Do the two methods agree? Fit K-means with your chosen K (use `nstart = 25`).

   (c) (6 pts) Apply **hierarchical clustering** using Euclidean distance and complete linkage. Plot the dendrogram, and cut it into the same number of clusters as in (b) with `cutree()`.

   (d) (4 pts) Fit a **Gaussian mixture model** with `Mclust()` from the `mclust` package, allowing 1 to 6 clusters. How many clusters does BIC choose?

   (e) (4 pts) Compare the K-means and hierarchical clusters with `table()`. Which states change clusters between the two methods? Describe each K-means cluster using its mean on the original (unscaled) variables, e.g., "high crime, mostly urban."

   (f) (2 pts) In two or three sentences, describe how these clusters could be used by a public health or public safety agency.

Setup for Q2–Q4: Diabetes in NHANES

Questions 2–4 use adults from the `NHANES` data set (in the `NHANES` package). The outcome is self-reported `Diabetes` (Yes/No), which about 11% of these adults have. Use the following code to create the data:

```
library(NHANES)
library(tidyverse)

nh <- NHANES |>
  distinct(ID, .keep_all = TRUE) |>
  filter(Age >= 20) |>
  select(Diabetes, Age, Gender, Race1, Poverty, BMI, BPSysAve,
         BPDiaAve, TotChol, DirectChol, PhysActive, Pulse,
         SleepHrsNight) |>
```

```
drop_na() |>
mutate(Diabetes = factor(Diabetes, levels = c("Yes", "No")))
```

Putting `"Yes"` first makes it the event of interest for the `yardstick` metrics.

2. **Class Imbalance and SMOTE [25 points]**

   (a) (3 pts) Report the sample size and the number and percent of adults with diabetes. Split the data into training (70%) and test (30%) sets, stratified on `Diabetes`.

   (b) (3 pts) Create a recipe that makes dummy variables for the categorical predictors and normalizes the numeric predictors. Create a second recipe that adds `step_smote(Diabetes)`, from the `themis` package, to the first one. Using `prep()` and `bake()`, show the number of Yes and No in the training data after the SMOTE recipe, and in the test data after the SMOTE recipe. Explain why they differ.

   (c) (8 pts) Fit four models on the training data: logistic regression and a random forest (`ranger`, 500 trees; no tuning needed), each with and without SMOTE. For each model, report on the test set: ROC-AUC, PR-AUC, sensitivity and specificity at the default 0.5 threshold, and the average predicted probability of diabetes. Put the results in one table, along with the observed prevalence in the test set.

   (d) (4 pts) Without SMOTE, why is sensitivity so low at the 0.5 threshold? What happens to sensitivity and specificity with SMOTE? What happens to the AUC?

   (e) (4 pts) Compare the average predicted probability with the observed prevalence. What does SMOTE do to the predicted risks, and why? Give one situation where this would be a problem and one where it would not.

   (f) (3 pts) A classmate applies SMOTE to the full data set and then splits it into training and test sets. Explain two reasons why their test-set results would be too optimistic.

3. **Interpretable Models [15 points]**

   (a) (5 pts) Using the same training data as in Q2 (without SMOTE), fit a classification tree with `rpart` and `control = rpart.control(maxdepth = 3, cp = 0.002)`. (With the default `cp = 0.01`, the tree does not split at all.) Plot the tree (e.g., with `rpart.plot`). Write out the path that leads to the leaf with the highest predicted risk as a rule in words, e.g., "If age $\geq \ldots$ and BMI $\geq \ldots$, the predicted risk is $\ldots$"

   (b) (3 pts) Look at the variables and the direction of each split in your tree. Does any split surprise you? Give a possible explanation. (Hint: think about what treatment people with diabetes often receive.)

(c) (3 pts) Compute the test-set ROC-AUC and PR-AUC of the tree, and compare it with the ROC-AUCs and PR-AUCs of the logistic regression and random forest from Q2 (without SMOTE). Give a clear indication what is lost by using the simpler models (if anything)?

(d) (4 pts) Rudin (2019) argues that for high-stakes decisions we should *"stop explaining black box machine learning models ... and use interpretable models instead."* In one paragraph, say whether your results support this argument for this problem. Which of the models in Q2 and Q3 would you choose to show a clinician, and why?

4. **Explainable AI [25 points]**

Use the random forest from Q2 **without** SMOTE for this question. Several of the functions below need a prediction function that returns the predicted probability of diabetes. If `rf_fit` is your fitted workflow, you can use:

```
pred_yes <- function(object, newdata) {
  predict(object, new_data = newdata, type = "prob")$.pred_Yes
}
```

To keep the computations fast, you may use a random sample of 500 training observations wherever the methods need "background" data.

(a) (5 pts) **Permutation importance.** Compute and plot permutation variable importance (e.g., with `vip::vi_permute()` or `vip::vip(method = "permute")`). Which three predictors are most important? Explain in one or two sentences what permutation importance measures.

(b) (6 pts) **Partial dependence.** Create partial dependence plots (PDPs) for `Age` and `BMI`. The simplest way is with the `iml` package, which works directly with `pred_yes`:

```
library(iml)
pred <- Predictor$new(rf_fit, data = background,
                      predict.function = pred_yes)
plot(FeatureEffect$new(pred, feature = "Age", method = "pdp"))
```

where `background` is a data frame of (a sample of) the training predictors. Describe the shape of each relationship. Is either one close to linear?

(c) (5 pts) **ALE.** `BPSysAve` (systolic blood pressure) is correlated with `Age`. Report this correlation, and then create both a PDP and an ALE plot for `BPSysAve` (`method = "ale"` in `FeatureEffect`). Do they tell the same story? Why can the PDP be misleading when predictors are correlated?

(d) (6 pts) **SHAP.** Choose the test-set adult with the highest predicted risk. Compute SHAP values for this person (e.g., with `fastshap::explain()`, `nsim = 50`) and

plot them. Which predictors push this person's predicted risk up the most, and do any push it down? Explain in plain language, as you would to the person's doctor.

(e) (3 pts) Which of the methods in (a)–(d) are **global** and which are **local**? Name one limitation of explaining the random forest this way rather than using an interpretable model like the one in Q3.

5. **Concepts [10 points]**

Answer each in two or three sentences.

(a) (3 pts) What is the difference between interpretable machine learning (IML) and explainable AI (XAI)? Give one example of each.

(b) (3 pts) A colleague says, "To get an interpretable model you always have to give up some accuracy." Do you agree? Use your results from Q3 in your answer.

(c) (4 pts) Give one public health example where an interpretable model is essential, and one where an accurate black box model (with XAI explanations) may be acceptable. Explain the difference.